Exam number	
Section	

Process and Operations Management
Final Exam – Solutions

Profs. Ilgin Dogan and Florin Ciocan

- Duration of the exam: 3 hours.
- There are **6 problems** in the exam. Read through all the questions first and start with the ones you feel the most comfortable with.
- This final is **open book**; this means you can use all your class materials, handouts and any other material that can reasonably be brought into the exam hall. Due to the individual nature of the exam, using any sort of device that has communication abilities is not allowed. This includes computers, phones in all form factors (including handhelds). Use of traditional scientific calculators is permitted to solve analytical portions of the exam, but the logic and steps should be clearly indicated in your written response. In case you need them, the tables are provided on the last pages.
- Your answers must be in English. Be **precise and to the point**. Short, clear, well thought out answers are preferred. **Neatness** will be rewarded!
- All answers must be provided on this exam paper, inside the boxes below each question.
- Some questions ask for quantitative answers. Even if you cannot obtain them, it may be worthwhile to provide the intuition or the expected business insight. Indicating what computations you might do (if you had more time, had all the data needed, etc.) may also be helpful.
- Make **reasonable** assumptions when you feel this is necessary in order to answer a question, but then clearly indicate that you did so and why you did so. Of course, the need for, and the appropriateness of the assumption will be evaluated by the grader.
- This exam consists of 27 printed pages.
- Good Luck!

Points							
Problem	1	2	3	4	5	6	Total
Your Score	/18	/18	/26	/19	/18	/26	/125

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

1. Two Paths in Second-Hand Fashion: Vinted vs. The RealReal [18 points]

Driven by sustainability trends, better tech, and shifts in consumer values, the second-hand fashion market took off in the early-to-mid 2010s and boomed after 2015. Within this growing space, two platforms—Vinted and The RealReal—have become well-known names, each offering a very different experience for buyers and sellers.

Vinted began in 2008 as a small startup in Lithuania and quickly expanded across Europe. Its model is simple: individuals list items directly on the platform, name their own prices, and ship goods to buyers once sold. There is no middle layer—no inspections, warehouses, or expert interventions. Sellers build their reputation through peer ratings, while buyers are protected by a small fee in case of non-delivery or defects. Vinted generates revenue from buyer commissions and optional paid features like promoted listings. CEO Thomas Plantenga notes, "I think our popularity comes from how easy it is to use Vinted and make money". This lean, peer-to-peer structure has enabled Vinted to achieve net profitability since 2023, making it a rare exception among major second-hand fashion marketplaces.

Meanwhile, The RealReal, founded in the United States in 2011, adopts a more hands-on approach. Sellers send in their items, which are received, authenticated, photographed, and stored in company-operated facilities. Experts evaluate each piece before it goes live on the platform. Once sold, The RealReal handles packaging and shipping. In return for this full-service consignment model, the company takes a commission from sellers, which varies based on several factors. This model offers a high degree of quality and curation. "We believe prioritizing authentication and transparency keeps customers coming back," explains Rati Levesque, CEO of The RealReal. Although the company has reported significant gross margin improvements and progress toward net profitability, it has not yet reached that milestone.

Despite all the hype of second-hand fashion, profiting from the shift to thrift has become challenging for both companies and their competitors. While both platforms promote sustainability and operate in the same space, their business models differ sharply and come with their own set of operational trade-offs.

QUESTIONS BEGIN ON THE NEXT PAGE.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(1a) Why might Vinted be profitable while The RealReal is not, even though The RealReal offers a more premium service?

[4 points]

The RealReal operates a much more hands-on, operationally intensive model. It handles end-to-end logistics, including warehousing, authentication, and photography of items, which adds significant labor and infrastructure costs. These operational risks and overhead make it harder to scale profitably.

In contrast, Vinted runs a peer-to-peer marketplace that's much more asset-light. It doesn't take on the same level of responsibility for handling goods, which keeps operational complexity low. Combined with high user volume due to lower transactional costs, this allows Vinted to achieve profitability more easily.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(1b) Compare and contrast the operational trade-offs of each marketplace across the dimensions highlighted in class: scalability and market thickness, trust and safety, price and value discovery, and transaction costs.

[8 points]

Trust & Safety vs. Transaction Costs:

- The RealReal prioritizes trust and strict quality control, which appeals to their relatively inelastic customer base that is willing to pay for a premium experience. However, this comes at the cost of high seller commissions and expensive transactions.
- Vinted keeps quality control minimal to reduce transaction costs. They crowdsource trust through seller ratings while maintaining operations lean. Although this approach results in variability in quality and potential trust issues, it aligns with the brand's narrative centered on affordability and works well for it elastic, budget-conscious customer pool.

Trust & Safety vs. Scalability:

- The RealReal's authentication-based business model is operationally complex and resource-intensive. The high transaction costs favor professional, high-priced sellers and make the platform less attractive to independent, low-volume sellers. Therefore, as the platform grows, it faces diminishing returns due to increasing inventory and logistical challenges, which ultimately constrain its speed and scalability.
- Vinted benefits from a decentralized and seller-managed operational model. By minimizing operational interventions and leveraging network effects, Vinted offers a simple and cheap marketplace, which in turn enables rapid growth of market thickness and high throughput. Yet this model sacrifices safety and oversight due to large reliance on variable seller behavior and limited buyer protections.

Price vs. Value Discovery:

- The RealReal ensures price consistency through expert- and data-driven pricing tailored to each brand. As a result, buyers get market-aligned prices and know what to expect. This offers a smooth experience and consistent price discovery. On the other hand, sellers have limited control over pricing and may feel constrained on their ability to influence value discovery or capture perceived worth.
- Vinted offers flexibility for value discovery by allowing sellers set their own prices and creating opportunities for price negotiation or bargains. However, this leads to inconsistent price discovery for buyers as they encounter a wide range of prices for similar items.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(1c) What operational strategies would you recommend for The RealReal to address its challenges in achieving profitable scalability? Support your answer with relevant operational concepts from the course, focusing on how the platform's process and product choices could better align with its brand narrative.

[6 points]

- *Focus (choice of product & customer segments):* Aligning with its brand focus on trust and rigorous authentication, The RealReal should concentrate exclusively on high-end luxury collections and target a relatively inelastic customer segment with a higher willingness to pay.
- *Splitting (for authentication process):* Use tiered authentication by appling AI or semi-automated checks for relatively mid-tier brand categories to reduce labor and speed up.
- *Splitting (for fulfillment process):*Adopt a hybrid fulfillment model by splitting logistics by brand category and using seller-managed shipping for select mid-tier products similar to Vinted's model.
- *Splitting (for commission scheme):* Implement a segmented commission structure to incentivize high-volume and/or high-value sellers.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

2. Supply Chain Contracts [18 points]

Firm A, which produces power management chips, enters a supplier relationship with Firm B. Firm B sells a new product line of servers that require Firm A's chips. Firm B sales team obtain a forecast of mean demand for their new server line. Since setting up the production line for Firm A's power chip requires completely new equipment, the firm needs to make an upfront capacity investment, and they rely on receiving the demand forecast from Firm B before they make their capacity decision.

(2a) Currently, Firm B buys the chips from A through a simple wholesale contract as seen in class. Use your intuition to explain what frictions may arise with such a setup, and how these could lower the profitability of the supply chain between Firms A and B.

[8 points]

In this wholesale setup, Firm A has limited visibility into actual market demand and must rely entirely on Firm B's forecasts. However, Firm B doesn't directly bear the irreversible cost of Firm A's capital investment if those forecasts turn out to be inaccurate.

This asymmetry creates significant incentive misalignment. Knowing that they don't directly pay for unused capacity, Firm B may strategically inflate forecasts to avoid the risk of stockouts and externalizing the risk to Firm A. While Firm B does face some forecast-related costs due to inventory mismatches, the burden on over-investment in capacity falls entirely on Firm A if the actual demand falls short. Further, if the demand turns out to be high, Firm A doesn't receive the corresponding benefit as its margin remains fixed at the wholesale price.

These misaligned incentives can lead to suboptimal capacity decisions. Either 1) Firm A may invest in less capacity than optimal due to lack of credibility with Firm B and result in lost sales for Firm B, or 2) Firm A may over-invest by trusting Firm B's inflated forecasts and incur losses due to wasted capacity. In either case, the supply chain fails to capture the full potential value of the new product launch and the overall profitability reduces.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(2b) One solution is to implement quantity flexibility contracts. Under such a contract, Firm B provides an initial forecast to Firm A but then has to purchase some quantity within a certain percentage, plus or minus, of this forecast. For example, let's say the firms agree to a 25% quantity flexibility contract in the first quarter of the year and Firm B forecasts demand will be 2000 units in the fourth quarter. By the time the fourth quarter comes, Firm B is committed to purchasing at least 1500 units, and Firm A is committed to delivering up to 2500 units.

Describe how this new contract can lead to a better coordinated supply chain. Pay attention to discuss what are the benefits and drawbacks of this contract.

[10 points]

Benefits:

Because Firm B's forecasts are now partially binding, it has less incentive to manipulate them. As a result, the quantity flexibility contract reduces the risk of moral hazard and enables Firm A to make better-informed, less risky, and more efficient capacity investment decisions.

This coordination shifts a portion of the mismatch risk from Firm A to Firm B. Since Firm B has greater market visibility, it is better positioned to bear this risk. As a result, Firm A can make capacity decisions closer to the optimal level, while Firm B retains some operational flexibility to respond to actual demand within a reasonable range. This improved alignment of incentives and risk-sharing makes both parties better off and leads to higher supply chain profitability compared to a simple wholesale contract.

Drawbacks:

- Firm A must still prepare for the upper bound (e.g., 2500 units) and continues to shoulder most of the overage risk. On top of that, they now have to deliver higher volumes on shorter notice, which can put pressure on both labor and procurement, especially if some components have long lead times.
- Firm B's retained flexibility can now lead to a more subtle strategic misbehavior. Instead of inflating forecasts, Firm B might now understate them. For example, they may choose forecasting and paying for 1500 units (when they actually expect 2000) and then ramp up orders to 2000 under the flexibility clause. This manipulation would still disrupt Firm A's capacity planning.
- The new setup gives Firm B some room to adjust to real demand, but provides limited protection against extreme market volatility. If real demand falls outside the agreed flexibility range (e.g., due to macroeconomic shifts) and there are no alternative buyers for the leftovers, the rigid contract terms can create both operational inefficiencies and contractual friction. In such cases, Firm B may hesitate to fully commit, and Firm A could struggle to enforce the contract.
- Another practical challenge arises from the negotiation complexity. Since uncertainty is not fully eliminated, setting the flexibility percentage requires a careful balance of risk, trust, and demand variability. Too much flexibility increases Firm A's risk, whereas too little restricts Firm B's reactivity, and these trade-offs typically makes it difficult for both parties to agree on a fair rate.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

3. Business Intuition [26 points]

Please circle the best answer for each multiple-choice question below. [3 points each]

A. Which of the following strategies help reduce information risk in a business model?

(i) Making all production and resource allocation decisions upfront to fully commit to demand forecasts.

(ii) Investing in dedicated capacity that is optimized for the average demand scenario.

(iii) Splitting decisions across time by deferring some commitments until more information is available.

(iv) Standardizing products and processes to reduce variability in supply chains.

a. (i) and (ii) only
b. (iii) only
c. (iii) and (iv) only
d. All of the above

Answer: b

B. A production process is experiencing delays due to a bottleneck stage that limits overall system capacity. Which of the following strategies would effectively alleviate the bottleneck?

(i) Increasing the capacity of the bottleneck stage by adding equipment or labor.

(ii) Adding a quality control step before the bottleneck to reduce defective items reaching it.

(iii) Adding a quality control step at the bottleneck to inspect items during the most constrained step.

(iv) Re-routing some of the work that goes through the bottleneck to an alternative parallel process.

(v) Increasing production rates at non-bottleneck stages to build buffer inventory and avoid starving the bottleneck.

a. (i), (ii), and (iv)
b. (i), (ii), and (v)
c. (i), (iii), and (iv)
d. (i), (iii), and (v)
e. (ii), (iii), and (iv)
f. All of the above

Answer: a

C. Which of the following statements about queuing analysis is NOT always true?

a. The goal is to achieve a balance between the cost of providing a service and the cost of customer waiting for that service.
b. The average number of customers in the system tends to increase when either the arrival rate increases or the time each customer spends in the system gets longer.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

c. Improving service capacity increases the throughput rate of the queue.

d. A stable queue requires that the average service rate exceeds the average arrival rate.

Answer: c

D. Which of the following statements about linear programming (LP) in the context of pricing and capacity allocation is misleading?

a. LP maximizes total revenue by allocating limited capacity across various customer segments based on opportunity costs.

b. It identifies the lowest-cost way to provide uniform prices to all customers regardless of their willingness to pay.

c. It scales to high-dimensional allocation problems, making it useful in complex industries like airlines or retail.

d. It enables clear decision rules that managers can trace and adjust based on evolving market conditions.

Answer: b

E. Decide whether the following comparisons between the Toyota Production System (TPS) and mass manufacturing are True (T) or False (F). [**2 points each**]

T / F TPS typically optimizes batch sizes to justify long setup times, whereas mass manufacturing aims to minimize setup times.

T / F TPS strives for high utilization of bottlenecks only, while mass manufacturing aims for high utilization across all resources.

T / F TPS reduces the bullwhip effect through frequent, small-lot production and pull systems, while mass manufacturing can amplify it due to forecast-driven push systems.

T / F TPS encourages negotiating with suppliers for lower material costs every month to improve profitability without a concern for quality or lead times.

Answers: F, T, T, F

F. A company is evaluating whether to invest in a flexible workforce that can switch between two product lines. This flexibility comes at an added cost of $5000 per week. Under which scenario would the investment in flexibility be most likely to pay off?

a. Weekly capacity is 1000 units, and demand is consistently 800 units.

b. Weekly capacity is 1000 units, and demand is consistently 1200 units.

c. Weekly capacity is 1000 units, and expected demand is also 1000 units, though actual demand can vary by ± 50 units.

d. Weekly capacity is 1000 units, and expected demand is also 1000 units, though actual demand can vary by ± 200 units.

Answer: d

G. In a service environment, which of the following statements best reflects a true consideration when comparing a pooled queue system to a system with separate queues for each server?

a. Pooling increases total waiting time because customers must wait in a single line regardless of which server becomes available.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

b. Pooling fully eliminates service time variability by spreading demand evenly across all servers.

c. Pooling would be most effective if service times across different customer types are positively correlated.

d. One potential caveat of pooling is that it may require additional investment in cross-training servers to handle multiple service types.

Answer: d

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

4. Beleza Natural [19 points]

You are hired as a consultant by the Beleza Natural Institute, a network of hair salons in Brazil with yearly revenues of $40 million. The institute's main hairdressing service is the Super-Relaxant (SR) treatment. This exclusive treatment uses a patented product that is applied from the roots to the ends of a client's hair to make it shinier and softer, form smooth waves and curls, reduce volume and create greater hair movement.

In contrast to classic hair salons, the Beleza Natural Institute specializes in one service and has low costs. In addition, the service is organized similarly to a production line. Customers arrive at a constant rate to the institute during the day. The process includes three stages. (1) Upon arrival, customers take a ticket with a number and make a payment. It takes 2 min for one worker to process the payment and give further instructions to the client. (2) Then, customers go through the division process - their hair is separated into small triangular sections. It takes a worker 12.5 min to process a single client. (3) Finally, the client is given the SR treatment itself. This stage takes 40 min for one worker.

Currently, each worker of the Beauty Institute works on one activity only. The Beauty Institute hired 2 workers for stage (1), 8 workers for stage (2), and 18 workers for stage (3). Assume that the time that each stage takes is deterministic and has no variability.

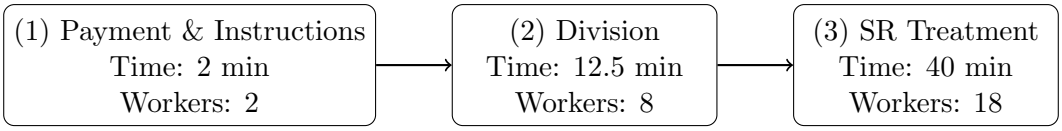
Due to rapid growth in the number of stores, the institute hired you to examine their process and potentially implement an operational improvement plan.

QUESTIONS BEGIN ON THE NEXT PAGE.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(4a) Draw a diagram of the process, showing the stages of the process, the time needed to accomplish a task by one worker, and the number of workers assigned to each stage. [3 points]

Solution:



(4b) Where is the bottleneck in the process and what is the system capacity? Analyze the capacity of each step of the process. [4 points]

Solution:

Capacity of each stage (customers per **hour**) is

$$\text{Capacity} = \frac{60}{\text{processing time (min/customer)}} \cdot (\#\text{ of workers}).$$

Stage (1):

$$\frac{60 \cdot 2}{2} = 60 \text{ customers/hour.}$$

Stage (2):

$$\frac{60 \cdot 8}{12.5} = \frac{480}{12.5} = 38.4 \text{ customers/hour.}$$

Stage (3):

$$\frac{60 \cdot 18}{40} = \frac{1080}{40} = 27 \text{ customers/hour.}$$

The bottleneck is the stage with the minimum capacity, which is Stage (3).
Thus, the system capacity is 27 customers per hour.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Suppose that customer demand depends on the day of the week. On Fridays, the Beleza Natural Institute receives 396 customer requests for appointments, on other working days, there are 288 customers per day requesting an appointment. Assume that there is no variability and that customers arrive uniformly spaced during the day. The institute is closed on Saturday and Sunday. Assume that the Institute is open for 12 hours per working day.

(4c) What is the maximum number of customers that the institute can handle per day? What is the utilization of each step in the process in this case? Also, is the system capacity enough to cover the customer requests on all days of the week?

[3 points]

Solution:

From part b, the system can process per day: $27 \times 12 = 324$ customers/day.

Demand:

- Friday: 396 customers.
- Each of Monday–Thursday: 288 customers.

The maximum number of customers that the institute can handle per day is the daily throughput of the entire system.

Throughput (i.e., flow rate) per day is $\min(\text{Demand}, \text{Capacity}) = \min(\text{Demand}, 324)$, so

- Friday throughput: $\min(396, 324) = 324$.
- Monday–Thursday throughput: $\min(288, 324) = 288$.

Based on that,

- Monday–Thursday: $288 \leq 324 \Rightarrow$ capacity is sufficient.
- Friday: $396 > 324 \Rightarrow$ capacity is not sufficient; $396 - 324 = 72$ customers cannot be served on Friday.

Daily available time per worker is $12 \times 60 = 720$ minutes.

Utilization of a stage is

$$\rho = \frac{\text{throughput}}{\text{capacity}} = \frac{\text{throughput} \cdot \text{processing time}}{(\#\text{ workers}) \cdot 720}.$$

On Fridays (when throughput = 324 customers):

Stage (1):

$$u_1 = \frac{324 \cdot 2}{2 \cdot 720} = 0.45.$$

Stage (2):

$$u_2 = \frac{324 \cdot 12.5}{8 \cdot 720} = \frac{4050}{5760} \approx 0.703.$$

Stage (3):

$$u_3 = \frac{324 \cdot 40}{18 \cdot 720} = \frac{12960}{12960} = 1.$$

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

On Mon-Thu (when throughput = 288 customers):

Stage (1):

$$u_1 = \frac{288 \cdot 2}{2 \cdot 720} = 0.40.$$

Stage (2):

$$u_2 = \frac{288 \cdot 12.5}{8 \cdot 720} = \frac{3600}{5760} = 0.625.$$

Stage (3):

$$u_3 = \frac{288 \cdot 40}{18 \cdot 720} = \frac{11520}{12960} \approx 0.889.$$

(4d) Compute the direct labor costs incurred for processing one customer. Suppose that hiring one worker costs \$500 per week. **Hint: Take into account that the customer flow rate varies throughout the week.**

[3 points]

Solution:

Total number of workers:

$$2 + 8 + 18 = 28.$$

Weekly cost (each worker costs \$500 per week):

$$28 \cdot 500 = 14000 \text{ dollars per week.}$$

Weekly number of customers actually processed (from the throughput calculations in part c):

$$4 \times 288 + 324 = 1152 + 324 = 1476.$$

Direct labor cost per customer:

$$\text{Cost per customer} = \frac{14000}{1476} \approx 9.49.$$

So, the direct labor cost is approximately \$9.49 per customer.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Consider a more complicated process, where stage (1) is the same as before: an incoming customer takes a ticket and makes the payment for a visit. It takes 2 min for a worker to process the payment and give further instructions to the client. However, at stage (2), customers are examined to find whether they are suitable for the Super-Relaxant treatment. It takes 5 min for the examination of one client by one worker. 3 specialists are assigned to this stage of the process. It turns out that, on average, 20% of clients are not suitable for the treatment. Those who are suitable, go through (3) Division and (4) SR treatment. Here everything is the same as above.

The Division process (3) takes 12.5 min for one worker to complete for a single client. The SR treatment (4) takes 40 min to be administered by one worker. 8 workers are assigned to stage (3), and 18 workers to stage (4).

For all customers, whether they are suitable for the treatment or not, the Beleza Natural Institute offers (5) Hydration and Hairstyle service, abbreviated with HH. A worker needs 10 min of time to administer HH to a client and 4 workers are assigned to this task. It turns out that clients who are treated with SR, also take the HH service in 95% of the cases, but those who are not suitable for the SR treatment only take the HH service in 50% of the cases.

(4e) Draw a diagram of the process, showing the stages of the process, the time needed to accomplish a task by one worker, and the number of workers assigned to each stage, as well as the percentages of clients going through each step of the process.

[3 points]

Solution:

Stages:

- (1) Payment: 2 min per customer, 2 workers.
- (2) Suitability exam: 5 min per customer, 3 workers.
- (3) Division: 12.5 min per customer, 8 workers.
- (4) SR treatment: 40 min per customer, 18 workers.
- (5) Hydration and Hairstyle (HH): 10 min per customer, 4 workers.

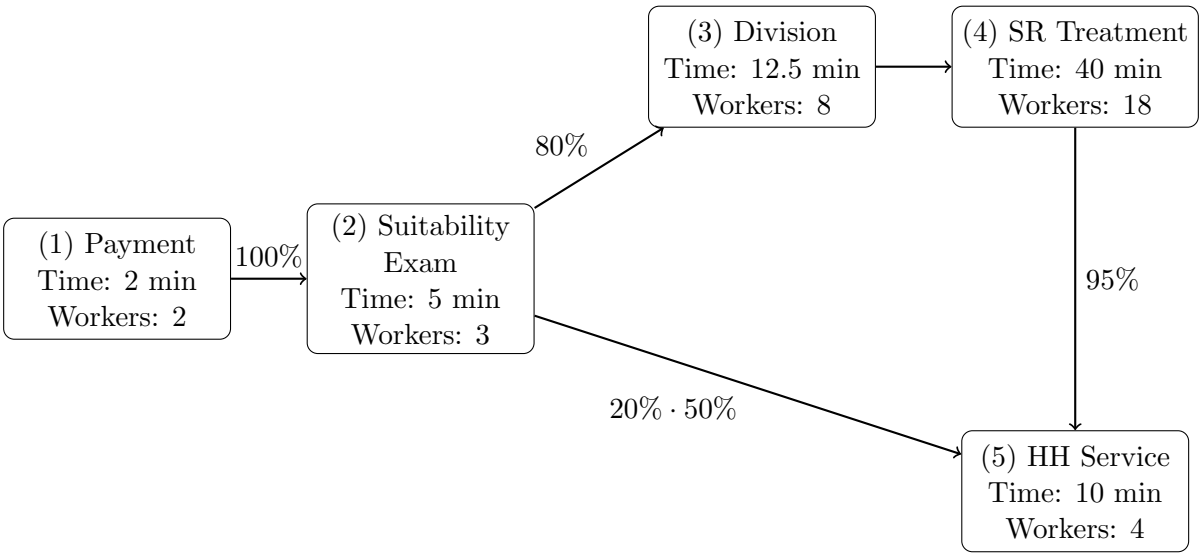
Routing:

- At stage (2), 80% of customers are suitable for SR, 20% are not.
- Suitable customers go through stages (3) and (4).
- All customers (suitable or not) may receive HH (stage (5)):
  - 95% of SR-treated customers receive HH.
  - 50% of non-suitable customers receive HH.

Overall fraction of customers using HH:

$$0.8 \cdot 0.95 + 0.2 \cdot 0.50 = 0.76 + 0.10 = 0.86.$$

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.



(4f) Assume now that there is always enough demand. For this new process, where is the bottleneck and what is the system capacity? Analyze the utilization of each step of the process if the bottleneck is fully utilized.

[3 points]

Solution:

Let R be the overall throughput (customers per hour entering stage (1)). Fractions visiting each stage:

- Stage (1): 1.00.
- Stage (2): 1.00.
- Stage (3): 0.80 (only suitable customers).
- Stage (4): 0.80.
- Stage (5) (HH): 0.86 (from the calculation above).

Ignoring routing fractions, capacities (per hour) of the resources are:

$$\begin{aligned}
\text{Stage 1:} \quad & \frac{60 \cdot 2}{2} = 60, \\
\text{Stage 2:} \quad & \frac{60 \cdot 3}{5} = 36, \\
\text{Stage 3:} \quad & \frac{60 \cdot 8}{12.5} = 38.4, \\
\text{Stage 4:} \quad & \frac{60 \cdot 18}{40} = 27, \\
\text{Stage 5:} \quad & \frac{60 \cdot 4}{10} = 24.
\end{aligned}$$

Convert these into constraints on R (total incoming flow):

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

$$\begin{aligned}
&\text{Stage 1: } R \le 60,\\
&\text{Stage 2: } R \le 36,\\
&\text{Stage 3: } 0.8R \le 38.4 \;\Rightarrow\; R \le 48,\\
&\text{Stage 4: } 0.8R \le 27 \;\Rightarrow\; R \le 33.75,\\
&\text{Stage 5: } 0.86R \le 24 \;\Rightarrow\; R \le \frac{24}{0.86} \approx 27.91.
\end{aligned}$$

The tightest bound is from Stage (5), so

- Bottleneck: Stage (5) HH.
- System capacity:
$$R^* = \frac{24}{0.86} \approx 27.91 \text{ customers/hour.}$$

Lastly, the utilization of each stage is calculated as follows.

Let f_i be the fraction of customers visiting stage i, t_i the processing time (minutes), and w_i the number of workers.

Utilization of a stage is

$$\begin{aligned}
\rho &= \frac{\text{throughput}}{\text{capacity}}\\
&= \frac{\text{throughput} \cdot \text{processing time} \cdot \text{fraction of customers entering}}{60 \cdot (\#\text{ workers})}\\
&= \frac{R^* \cdot f_i \cdot t_i}{60 \cdot w_i}.
\end{aligned}$$

Using $R^* \approx 27.91$:

- Stage (1): $f_1 = 1$, $t_1 = 2$, $w_1 = 2$,
$$u_1 = \frac{27.91 \cdot 1 \cdot 2}{60 \cdot 2} \approx 0.465.$$
- Stage (2): $f_2 = 1$, $t_2 = 5$, $w_2 = 3$,
$$u_2 = \frac{27.91 \cdot 1 \cdot 5}{60 \cdot 3} \approx 0.775.$$
- Stage (3): $f_3 = 0.8$, $t_3 = 12.5$, $w_3 = 8$,
$$u_3 = \frac{27.91 \cdot 0.8 \cdot 12.5}{60 \cdot 8} \approx 0.581.$$
- Stage (4): $f_4 = 0.8$, $t_4 = 40$, $w_4 = 18$,
$$u_4 = \frac{27.91 \cdot 0.8 \cdot 40}{60 \cdot 18} \approx 0.827.$$
- Stage (5): $f_5 = 0.86$, $t_5 = 10$, $w_5 = 4$,
$$u_5 = \frac{27.91 \cdot 0.86 \cdot 10}{60 \cdot 4} = 1.$$

Thus, in the new process, the HH stage is the bottleneck with capacity about 27.91 customers per hour and 100% utilization.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

5. Phoenix For Rescue [18 points]

To improve customer experience and reduce pressure on human agents, a team of Instead MBAs led by Florencia developed Phoenix, an AI-powered chatbot designed to handle customer service queries.

Phoenix is programmed to engage in helpful, conversational interactions with customers, offering support on topics such as product troubleshooting, billing questions, and delivery updates. It is available to chat with customers via text messaging between 8 a.m. and 6 p.m. Eastern Time.

Customers interact with Phoenix through a shared virtual queue. When a customer initiates a chat, they are connected to the next available instance of Phoenix on a first-come, first-served basis. On average, each chat lasts about 8 minutes and is exponentially distributed. Based on current usage patterns, customer chat requests arrive randomly according to a Poisson process at a rate of 6 customers per hour.

Although Phoenix is cloud-based, each instance consumes a fixed amount of computing resources. To manage costs, the team has set a cap on the number of Phoenix instances that can run simultaneously.

Note: Show all calculations clearly. State any assumptions you make.

QUESTIONS BEGIN ON THE NEXT PAGE.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(5a) What is the minimum number of Phoenix instances that Florencia must deploy to ensure that the average waiting time to start a chat is less than one minute?

Show all your work clearly.

[6 points]

Solution:

Parameters:

$$\lambda = 6 \text{ customers/hr}, \mu = 8 \text{ mins/customer} = 0.13 \text{ hr/customer}, CV_A = CV_S = 1$$

We enumerate possible number of instances:

<u>$s = 1$</u>:

Utilization:

$$\rho = \frac{\lambda}{\frac{1}{\mu}} = \frac{6}{\frac{1}{8/60}} = 0.8$$

From Kingman's formula:

$$W_q = \mu \cdot \frac{\rho}{1-\rho} \cdot \frac{CV_A^2 + CV_s^2}{2} = 8 \cdot \frac{0.8}{0.2} \cdot 1 = 32 \text{ minutes.}$$

<u>$s = 2$</u>:

Utilization per Phoenix instance:

$$\rho = \frac{\lambda}{s \cdot \frac{1}{\mu}} = 0.4$$

From Kingman's formula:

$$W_q = \frac{\mu}{s} \cdot \frac{\rho^{\sqrt{2(s+1)}-1}}{1-\rho} \cdot \frac{CV_A^2 + CV_s^2}{2} = 1.77 \text{ minutes.}$$

<u>$s = 3$</u>:

Utilization per Phoenix instance:

$$\rho = \frac{\lambda}{s \cdot \frac{1}{\mu}} = \frac{0.8}{3} = 0.2667$$

From Kingman's formula:

$$W_q = \frac{\mu}{s} \cdot \frac{\rho^{\sqrt{2(s+1)}-1}}{1-\rho} \cdot \frac{CV_A^2 + CV_s^2}{2} = 2.667 \cdot 0.1217 \cdot 1 = 0.325 \text{ minutes.}$$

So, with 3 Phoenix instances, we meet the target service level.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Assume Florencia and the team are now considering two strategies to improve performance and engagement.

The first strategy involves deploying three Phoenix instances and enhancing Phoenix's conversational tone to make it more engaging and empathetic. As a result, the average duration of a chat is expected to increase from 8 to 16 minutes (as Phoenix takes more time to craft thoughtful responses), but the number of incoming chats is also expected to increase by 50%.

The second strategy focuses on expanding Phoenix's support capabilities by integrating it with more internal tools and databases. Since this is more costly, they can only deploy one Phoenix instance in this case. This added functionality is expected to change the service times such that it will not be exponentially distributed. The average duration for a chat will be 8 minutes and the standard deviation of the duration will be 4 minutes.

(5b) Which of these strategies will lead to (i) higher average time spent in the queue for each customer; (ii) higher average time spent in the system for each customer. Provide an intuitive interpretation of the results based on your calculations for each strategy.

Show all your work clearly. [8 points]

Solution:

First Strategy:

$$\lambda = 9 \text{ customers/hr}, \mu = 16 \text{ mins/customer} = 0.27 \text{ hr/customer}, s = 3, CV_A = CV_S = 1$$

Then,

$$\rho = \frac{\lambda}{s \cdot \frac{1}{\mu}} = 0.81$$

From Kingman's formula:

$$W_q = \frac{\mu}{s} \cdot \frac{\rho^{\sqrt{2(s+1)}-1}}{1-\rho} \cdot \frac{CV_A^2 + CV_s^2}{2} = \frac{16}{3} \cdot \frac{0.81^{\sqrt{8}-1}}{0.19} \cdot 1 = 17.7467 \text{ minutes.}$$

$$W = W_q + \mu = 17.7467 + 16 = 33.7467$$

Second Strategy:

$$\lambda = 6 \text{ cus/hr}, \mu = 8 \text{ mins/cus} = 0.13 \text{ hr/cus}, s = 1, CV_A = 1, CV_S = \frac{4}{8} = 0.5$$

Then,

$$\rho = \frac{\lambda}{s \cdot \frac{1}{\mu}} = 0.8$$

From Kingman's formula:

$$W_q = \mu \cdot \frac{\rho}{1-\rho} \cdot \frac{CV_A^2 + CV_s^2}{2} = 8 \cdot \frac{0.8}{0.2} \cdot \frac{1.25}{2} = 20 \text{ minutes.}$$

$$W = W_q + \mu = 20 + 8 = 28$$

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Under the second strategy, the average time spent in the queue is longer, but a customer spends less time in the system compared to the first strategy. This is because although there is lower variation under the second strategy, there is only one server; thus customers must wait. Under the first strategy, there is more variation and longer service times, which results in a longer time spent in the system, but having multiple servers is helpful to reduce the wait.

It is expected that after the first year, a new and improved interface will allow each Phoenix instance to handle up to two customer conversations simultaneously, without any significant delay in replies. That is, the average duration of a chat will still be 8 minutes, even when Phoenix is serving two customers at a time.

By that time, the subscriber base is projected to double, leading to a doubling of the chat arrival rate—from 6 to 12 customer chats per hour.

(5c) How many Phoenix instances will Florencia need to deploy to ensure that the average waiting time in the queue remains below one minute?

Show all your work clearly. [4 points]

Solution:

$$\lambda = 12 \text{ customers/hr}, \mu = 8 \text{ mins/customer} = 0.13 \text{ hr/customer}, CV_A = CV_S = 1$$

We enumerate possible number of instances:

<u>One Phoenix instance $\Rightarrow s = 2$</u>:

Utilization:

$$\rho = \frac{\lambda}{s \cdot \frac{1}{\mu}} = 0.8$$

From Kingman's formula:

$$W_q = \frac{\mu}{s} \cdot \frac{\rho^{\sqrt{2(s+1)}-1}}{1-\rho} \cdot \frac{CV_A^2 + CV_s^2}{2} = 4 \cdot \frac{0.8^{\sqrt{6}-1}}{0.2} \cdot 1 = 10.49 \text{ minutes.}$$

<u>Two Phoenix instances $\Rightarrow s = 4$</u>:

Utilization per Phoenix instance:

$$\rho = \frac{\lambda}{s \cdot \frac{1}{\mu}} = 0.4$$

From Kingman's formula:

$$W_q = 2 \cdot \frac{0.4^{\sqrt{10}-1}}{0.6} \cdot 1 = 0.459 \text{ minutes.}$$

So, the new interface helps us achieving the target service level with 2 Phoenix instances.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

6. EduJoy [26 points]

EduJoy is a small social enterprise launched on Kickstarter three years ago. Its first product—a hands-on educational kit designed to teach basic science concepts through creative play—was an instant hit and earned several awards for social impact. EduJoy's sales peak around Christmas, especially as many customers purchase kits to donate to schools and non-profits. To plan for this high-impact period, EduJoy hired a team of consultants who estimated demand for the upcoming Christmas season to be normally distributed, with a mean of 20,000 kits and a standard deviation of 5,000.

EduJoy partners with two community-driven workshops abroad to produce the kit's key components: (a) a durable cardboard activity box and (b) a set of learning tools (such as small magnifying lenses, measurement cards, and wooden parts). These components are ordered two months in advance. Each kit requires one box and two sets of learning tools. The boxes are provided by CraftMakers Collective, while the tools come from LearnTools Co-op. CraftMakers charges \$4 per box, and LearnTools charges \$10 per tool set (two are needed per kit). The production costs for CraftMakers and LearnTools are \$1 per box and \$4 per tool set, respectively.

Once all components (1 box + 2 tool sets) are received, EduJoy assembles the kits at a local community center at a cost of \$3 per kit. Since the work is done in-house, assembly happens only after the real demand is known, i.e., only after observing the actual demand. For example, if demand is 200 kits but enough materials for 300 were ordered, only 200 kits will be assembled, so the total assembly cost would be 200 x \$3. The kits are sold at \$60, and any unused components can be returned to the workshops at 10% of their original purchase price (i.e., the suppliers buy back leftover parts at 10% of the original cost).

(6a) Find the optimal quantity of each kit component EduJoy should order before the Christmas season from its suppliers, i.e., number of boxes and number of learning tool sets.

Show all your work clearly. [4 points]

Solution:

Procurement cost: $c = 4 + 2 \times 10 = 24$
Salvage value: $s = 24 \times 0.1 = 2.4$
Price: $r = 60$

Then,
$c_u = 60 - (24 + 3) = \$33$
$c_o = 24 - 2.4 = \$21.6$

Critical ratio $= 0.604396$
$\Rightarrow$ From Standard Normal Table, $z^* = 0.27$
$\Rightarrow Q^* = \mu + z^* \times \sigma = 20,000 + 0.27 \times 5,000 = 21,350$ kits
$\Rightarrow 21,350$ boxes and $42,700$ learning tool sets.

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(6b) Calculate the current level of lost sales of EduJoy.

Show all your work clearly. [2 points]

Solution:

Using the Standard Lost Sales Table: $L(z^*) = L(0.27) = 0.27840$

Expected Lost Sales $= \sigma \times L(0.27) = 1392$ kits.

(6c) Calculate the profits of each supplier and EduJoy in this arrangement.

Show all your work clearly. [6 points]

Solution:

<u>EduJoy:</u>

Expected Sales $= \mu -$ Expected Lost Sales $= 20,000 - 1392 = 18,608$
Expected Leftover $= Q^* -$ Expected Sales $= 21,350 - 18,608 = 2742$
Expected Profit $= (60 - 3) \times 20,000 - 33 \times 1392 - 21.6 \times 2742 = \$1,034,836.8$

<u>Craftmakers:</u>

Expected Profit $= (4 - 1) \times 21,350 - 0.4 \times 2742 = \$62,953.2$

<u>LearnTools:</u>

Expected Profit $= (10 - 4) \times 42,700 - 1 \times 2 \times 2742 = \$250,716$

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Disappointed with the high level of lost sales, EduJoy decides to produce the two kit components also locally. If EduJoy chooses this option, it must pay \$6 for each activity box and \$20 for each set of learning tools. These components can then be produced and assembled (the cost of assembly remains \$3 per kit) fast enough to fulfill any unmet demand if needed.

(6d) Calculate the optimal number of activity box and learning tools ordered from abroad provided that the option of local production is feasible as well.

Show all your work clearly. [4 points]

Solution:

Cost of local production: $c_L = 6 + 2 \times 20 + 3 = \$49$.

Then,
$c_o = 24 - 2.4 = \$21.6$
$c_u = 49 - (24 + 3) = \$22$

Critical ratio $= 0.504587$
$\Rightarrow$ From Standard Normal Table, $z^* = 0.02$
$\Rightarrow Q^* = \mu + z^* \times \sigma = 20,100$ kits
$\Rightarrow 20,100$ boxes and $40,200$ learning tool sets.

(6e) Calculate the number of educational kits that will be produced locally provided that the option of local production is feasible.

Show all your work clearly. [2 points]

Solution: Expected Lost Sales $= \sigma \times L(0.02) \approx 1945$

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

(6f) Calculate EduJoy's profit after starting using the local manufacturer on top of sourcing from abroad. Compare it with your answer in part (6c) and advise whether EduJoy should use local production or not.

Show all your work clearly. [2 points]

Solution:
Expected Sales $= \mu -$ Expected Lost Sales $= 20,000 - 1945 = 18,055$
Expected Leftover $= Q^* -$ Expected Sales $= 20,100 - 18,055 = 2045.1$
Expected Profit $= (60-3) \times 20,000 - 22 \times 1945 - 21.6 \times 2045.1 = \$1,053,035.84$

Reactive capacity improves expected profits by approximately %2. So, EduJoy should use local manufacturer.

(6g) Consider the two scenarios below. Rethink the overage and underage costs. Explain how these would change.

Scenario 1: EduJoy is considering launching a second product—a "science extension pack"—which can be built from unsold boxes and tool sets remaining after the Christmas season. Each extension pack requires 1 leftover box and 2 leftover tool sets, and can be sold for \$20 (with no additional assembly cost).

Scenario 2: EduJoy is considering the local production option presented in part (6d). However, they learn that the local production is limited to 5000 kits and they cannot produce more than this amount.

Show all your work clearly. [6 points]

Answer:

<u>Scenario 1:</u>

c_u is the same.
$c_o = (24+3) - 20 = \$7$ (Here, we assume EduJoy assembles all the parts ordered, as they will be selling them (as a different product) at a discounted price.)

<u>Scenario 2:</u>

c_o is the same.

c_u depends on the order quantity from abroad. Say this quantity is Q^*, then the total supply available becomes at most $Q^* + 5000$

- If $Q^* < D \leq Q^* + 5000$, this is what we have already done in part d, so $c_u = 22$.

- If $D > Q^* + 5000$, then we will have lost sales that we could not even meet by local production. In this case, the underage cost would be $c_u = 33$ as in part a.

Suggested answer to this question ends here.

Below is a note to show how to incorporate this structure into the expected profit function to solve for the new optimal order quantity:

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Recall the definitions from Session 6:

Lost Sales $= \min(0, D - Q)$

Leftovers $= \min(0, Q - D)$

Then,

$$\begin{aligned}\text{Exp Profit} = &(r - c_{\text{abroad}}) * \mathbb{E}[D] \\ &- (c_{\text{local}} - c_{\text{abroad}}) * \mathbb{E}[\min(0, D - Q) - \min(0, D - Q - 50000)] \\ &- (r - c_{\text{abroad}}) * \mathbb{E}[\min(0, D - Q - 5000)] \\ &- c_o * \mathbb{E}[\min(0, Q - D)]\end{aligned}$$

Plugging in the numbers,

$$\begin{aligned}\text{Exp Profit} = &33 * \mathbb{E}[D] \\ &- 22 * \mathbb{E}[\min(0, D - Q) - \min(0, D - Q - 50000)] \\ &- 33 * \mathbb{E}[\min(0, D - Q - 5000)] \\ &- 21.6 * \mathbb{E}[\min(0, Q - D)]\end{aligned}$$

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Table A: Standardized Normal Probability Table

Z	0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
-2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
-2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
-2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
-2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
-2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
-2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
-2.3	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
-2.2	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
-2.1	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
-2.0	0.02275	0.02222	0.02169	0.02118	0.02068	0.02018	0.01970	0.01923	0.01876	0.01831
-1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
-1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
-1.7	0.04457	0.04363	0.04272	0.04182	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
-1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
-1.5	0.06681	0.06552	0.06426	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
-1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09853
-1.1	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.0	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.9	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
-0.8	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.7	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22363	0.22065	0.21770	0.21476
-0.6	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
-0.5	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
-0.4	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
-0.3	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
-0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
0.0	0.50000	0.49601	0.49202	0.48803	0.48405	0.48006	0.47608	0.47210	0.46812	0.46414
z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Table B: Standardized Normal Loss Table

z	0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
-2.9	2.90054	2.91052	2.92051	2.93049	2.94047	2.95046	2.96044	2.97042	2.98041	2.99040
-2.8	2.80076	2.81074	2.82071	2.83069	2.84066	2.85064	2.86062	2.87060	2.88058	2.89056
-2.7	2.70106	2.71103	2.72099	2.73096	2.74093	2.75090	2.76087	2.77084	2.78081	2.79079
-2.6	2.60146	2.61142	2.62137	2.63133	2.64129	2.65125	2.66121	2.67117	2.68113	2.69110
-2.5	2.50200	2.51194	2.52188	2.53183	2.54177	2.55171	2.56166	2.57161	2.58156	2.59151
-2.4	2.40272	2.41264	2.42256	2.43248	2.44241	2.45234	2.46227	2.47220	2.48213	2.49207
-2.3	2.30366	2.31356	2.32345	2.33335	2.34325	2.35316	2.36307	2.37298	2.38289	2.39280
-2.2	2.20489	2.21475	2.22462	2.23449	2.24436	2.25423	2.26411	2.27400	2.28388	2.29377
-2.1	2.10647	2.11629	2.12612	2.13595	2.14579	2.15563	2.16547	2.17532	2.18517	2.19503
-2.0	2.00849	2.01827	2.02805	2.03783	2.04762	2.05742	2.06722	2.07702	2.08683	2.09665
-1.9	1.91105	1.92077	1.93049	1.94022	1.94996	1.95970	1.96945	1.97920	1.98896	1.99872
-1.8	1.81428	1.82392	1.83357	1.84323	1.85290	1.86257	1.87226	1.88195	1.89164	1.90134
-1.7	1.71829	1.72785	1.73742	1.74699	1.75658	1.76617	1.77578	1.78539	1.79501	1.80464
-1.6	1.62324	1.63270	1.64217	1.65165	1.66114	1.67064	1.68015	1.68967	1.69920	1.70874
-1.5	1.52931	1.53865	1.54800	1.55736	1.56674	1.57612	1.58552	1.59494	1.60436	1.61380
-1.4	1.43667	1.44587	1.45508	1.46431	1.47356	1.48281	1.49208	1.50137	1.51067	1.51998
-1.3	1.34553	1.35457	1.36363	1.37270	1.38179	1.39090	1.40002	1.40916	1.41831	1.42748
-1.2	1.25610	1.26496	1.27384	1.28274	1.29165	1.30059	1.30954	1.31851	1.32750	1.33650
-1.1	1.16862	1.17727	1.18595	1.19465	1.20336	1.21210	1.22086	1.22964	1.23844	1.24726
-1.0	1.08332	1.09174	1.10019	1.10866	1.11716	1.12568	1.13422	1.14279	1.15138	1.15999
-0.9	1.00043	1.00860	1.01680	1.02503	1.03328	1.04156	1.04986	1.05819	1.06654	1.07491
-0.8	0.92021	0.92810	0.93603	0.94398	0.95196	0.95997	0.96801	0.97607	0.98417	0.99229
-0.7	0.84288	0.85048	0.85810	0.86576	0.87345	0.88117	0.88892	0.89669	0.90450	0.91234
-0.6	0.76867	0.77595	0.78325	0.79059	0.79797	0.80537	0.81281	0.82028	0.82778	0.83531
-0.5	0.69780	0.70473	0.71170	0.71870	0.72573	0.73281	0.73991	0.74705	0.75422	0.76143
-0.4	0.63044	0.63701	0.64362	0.65027	0.65695	0.66367	0.67042	0.67721	0.68404	0.69090
-0.3	0.56676	0.57296	0.57920	0.58547	0.59178	0.59813	0.60452	0.61094	0.61740	0.62390
-0.2	0.50689	0.51271	0.51856	0.52445	0.53038	0.53634	0.54235	0.54840	0.55448	0.56060
-0.1	0.45094	0.45635	0.46181	0.46731	0.47285	0.47842	0.48404	0.48969	0.49539	0.50112
0.0	0.39894	0.40396	0.40902	0.41412	0.41926	0.42444	0.42966	0.43492	0.44022	0.44556
z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.39894	0.39396	0.38902	0.38412	0.37926	0.37444	0.36966	0.36492	0.36022	0.35556
0.1	0.35094	0.34635	0.34181	0.33731	0.33285	0.32842	0.32404	0.31969	0.31539	0.31112
0.2	0.30689	0.30271	0.29856	0.29445	0.29038	0.28634	0.28235	0.27840	0.27448	0.27060
0.3	0.26676	0.26296	0.25920	0.25547	0.25178	0.24813	0.24452	0.24094	0.23740	0.23390
0.4	0.23044	0.22701	0.22362	0.22027	0.21695	0.21367	0.21042	0.20721	0.20404	0.20090
0.5	0.19780	0.19473	0.19170	0.18870	0.18573	0.18281	0.17991	0.17705	0.17422	0.17143
0.6	0.16867	0.16595	0.16325	0.16059	0.15797	0.15537	0.15281	0.15028	0.14778	0.14531
0.7	0.14288	0.14048	0.13810	0.13576	0.13345	0.13117	0.12892	0.12669	0.12450	0.12234
0.8	0.12021	0.11810	0.11603	0.11398	0.11196	0.10997	0.10801	0.10607	0.10417	0.10229
0.9	0.10043	0.09860	0.09680	0.09503	0.09328	0.09156	0.08986	0.08819	0.08654	0.08491
1.0	0.08332	0.08174	0.08019	0.07866	0.07716	0.07568	0.07422	0.07279	0.07138	0.06999
1.1	0.06862	0.06727	0.06595	0.06465	0.06336	0.06210	0.06086	0.05964	0.05844	0.05726
1.2	0.05610	0.05496	0.05384	0.05274	0.05165	0.05059	0.04954	0.04851	0.04750	0.04650
1.3	0.04553	0.04457	0.04363	0.04270	0.04179	0.04090	0.04002	0.03916	0.03831	0.03748
1.4	0.03667	0.03587	0.03508	0.03431	0.03356	0.03281	0.03208	0.03137	0.03067	0.02998
1.5	0.02931	0.02865	0.02800	0.02736	0.02674	0.02612	0.02552	0.02494	0.02436	0.02380
1.6	0.02324	0.02270	0.02217	0.02165	0.02114	0.02064	0.02015	0.01967	0.01920	0.01874
1.7	0.01829	0.01785	0.01742	0.01699	0.01658	0.01617	0.01578	0.01539	0.01501	0.01464
1.8	0.01428	0.01392	0.01357	0.01323	0.01290	0.01257	0.01226	0.01195	0.01164	0.01134
1.9	0.01105	0.01077	0.01049	0.01022	0.00996	0.00970	0.00945	0.00920	0.00896	0.00872
2.0	0.00849	0.00827	0.00805	0.00783	0.00762	0.00742	0.00722	0.00702	0.00683	0.00665
2.1	0.00647	0.00629	0.00612	0.00595	0.00579	0.00563	0.00547	0.00532	0.00517	0.00503
2.2	0.00489	0.00475	0.00462	0.00449	0.00436	0.00423	0.00411	0.00400	0.00388	0.00377
2.3	0.00366	0.00356	0.00345	0.00335	0.00325	0.00316	0.00307	0.00298	0.00289	0.00280
2.4	0.00272	0.00264	0.00256	0.00248	0.00241	0.00234	0.00227	0.00220	0.00213	0.00207
2.5	0.00200	0.00194	0.00188	0.00183	0.00177	0.00171	0.00166	0.00161	0.00156	0.00151
2.6	0.00146	0.00142	0.00137	0.00133	0.00129	0.00125	0.00121	0.00117	0.00113	0.00110
2.7	0.00106	0.00103	0.00099	0.00096	0.00093	0.00090	0.00087	0.00084	0.00081	0.00079
2.8	0.00076	0.00074	0.00071	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056
2.9	0.00054	0.00052	0.00051	0.00049	0.00047	0.00046	0.00044	0.00042	0.00041	0.00040

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.

Table C: L_q Values for the Multi-server Queue

Value of L_q for s servers, with mean utilization rate ρ, assuming Poisson arrivals and exponential service times (known as a $M/M/s$ queue).

Utilization rate ρ	**Number of servers (s)**				
	1	2	3	4	5
.10	.0111	.0020	.0004	.0001	.0000
.20	.0500	.0167	.0062	.0024	.0010
.30	.1286	.0593	.0300	.0159	.0086
.35	.1885	.0977	.0552	.0325	.0196
.40	.2667	.1524	.0941	.0605	.0398
.45	.3682	.2285	.1522	.1052	.0743
.50	.5000	.3333	.2368	.1739	.1304
.55	.6722	.4771	.3583	.2772	.2185
.60	.9000	.6750	.5321	.4306	.3542
.62	1.0116	.7743	.6213	.5109	.4269
.64	1.1378	.8880	.7246	.6051	.5130
.66	1.2812	1.0188	.8446	.7158	.6152
.68	1.4450	1.1698	.9847	.8461	.7367
.70	1.6333	1.3451	1.1488	1.0002	.8816
.72	1.8514	1.5500	1.3423	1.1834	1.0553
.74	2.1062	1.7914	1.5721	1.4025	1.2646
.76	2.4067	2.0785	1.8472	1.6668	1.5187
.78	2.7655	2.4237	2.1803	1.9887	1.8302
.80	3.2000	2.8444	2.5888	2.3857	2.2165
.82	3.7356	3.3661	3.0979	2.8832	2.7029
.84	4.4100	4.0265	3.7456	3.5190	3.3273
.86	5.2829	4.8852	4.5914	4.3526	4.1493
.88	6.4533	6.0414	5.7345	5.3834	5.2682
.90	8.1000	7.6737	7.3535	7.0898	6.8624
.92	10.5800	10.1392	9.8056	9.5290	9.2893
.94	14.7267	14.2712	13.9240	13.6344	13.3821
.96	23.0400	22.5698	22.2088	21.9060	21.6408
.98	48.0200	47.5350	47.1602	46.8439	46.5656
.99	98.0101	97.5176	97.1357	96.8127	96.4274

When one is faced with a value of ρ that lies between two values in the table, one may have to interpolate between the two values of L_q associated with the two successive values of ρ. For example, consider $s = 2$ with $\rho = 0.75$. This utilization rate lies in between the two table values of 0.74 and 0.76. The L_q for this $M/M/2$ queue can be approximated by the following linear interpolation:

$$L_q = 1.7914 + \frac{(2.0785 - 1.7914)}{(0.76 - 0.74)}(0.75 - 0.74) = 1.7914 + 0.1435 = 1.9349$$

This document is authorised for use only in MBA - Process & Operations Management (000149) at INSEAD - Mar 2026 - Sep 2026 – by Professor(s) Ville Satopaa. Copying, printing or posting is a copyright infringement.